

36th Maintenance Group Frequency Converters Diagnostic and Report (Hangars 2 & 5)

Statement of Work (SOW)

Description

The 36th Maintenance Group (MXG) requires a contractor to provide a comprehensive diagnostic and repair solution for the frequency converters (FCX) located in Hangars 2 and 5. The contractor shall perform an onsite Initial Visual Inspection and diagnostic evaluation to determine each unit's condition, and based on these findings, deliver a written report outlining the required repairs. As part of this evaluation, the contractor shall also verify and certify that the output signal quality complies with all aircraft power requirements and FCX performance specifications. Following certification, the contractor shall conduct aircraft-connected testing to confirm the aircraft's acceptance of the power. All work shall be performed in accordance with OEM technical guidance, industry standards, and applicable Air Force instructions.

Preferred Brand/Model Number

The required equipment for this effort is FCX frequency converter systems. All work shall be performed by technicians qualified and authorized to service FCX equipment.

1.0 Scope of Work

The following phased approach will be executed to diagnose the unit's condition and perform necessary reports.

Phase	Task	Description
1	Initial Visual Inspection	A thorough visual inspection of the converter unit will be performed to identify any visible signs of physical damage. This is based on prior information suggesting potential damage occurred during previous work.
2	Power Supply Verification	The voltage supply to the converter will be checked. The unit will then be powered on to measure the internal power supply voltage readings to ensure they meet specifications.

3	Buss Voltage Diagnostic	The main buss fuse will be temporarily removed, and the unit will be started to verify the correct buss voltage is present. This isolates the buss for accurate measurement.
4	Manual Mode Output Test	Following a successful buss voltage test, the fuse will be re-installed. The unit will be placed in manual mode and started. The output voltage will be increased incrementally to determine if the correct output can be achieved.
5	Component-Level Troubleshooting	If a voltage imbalance is detected during the manual mode test, the output capacitors will be disconnected and tested individually to isolate any faulty components.

2.0 Contingency for Circuit Board Failure

- In the event a specific circuit board is suspected of failure, boards from a secondary converter unit will be utilized for cross-testing to confirm the diagnosis.
- If it is determined that both the primary and secondary units have the same non-functional circuit boards, the troubleshooting phase of this SOW will conclude for that component.

3.0 Deliverables

Upon completion of the scope of work, the following will be delivered:

A single, detailed report that consolidates all findings, test results, and recommendations. This report shall be structured to include, at a minimum, the following sections for each frequency converter unit (Hangars 2 & 5):

- A high-level overview of the unit's overall condition.
- A definitive statement on the root cause(s) of failure.
- Proposed Solution: A clear recommendation for each unit (e.g., repair of specific components, full unit replacement).
- Repair/Replacement Timeline: A detailed schedule outlining the projected timeline to fully execute the proposed solution and return both units to a fully operational status.

- **Parts Analysis & Lead Time:** A list of all required parts, including part numbers. This section must state whether parts are commercially available or require special manufacturing, and provide the associated lead time for procurement.
- **Warranty Information:** A statement detailing the warranty period and terms for all proposed new components and for the contractor's workmanship.

4.0 Official Quotation for Recommended Work

The contractor shall provide a formal, itemized quotation detailing the full cost to implement the recommended work outlined in the Diagnostic Report.

This quotation must include a comparative option for full replacement if the total estimated repair cost for a unit is projected to exceed the cost of purchasing a new, equivalent unit. This comparative analysis shall present two distinct and separate options:

Option A: Full Repair Cost

A detailed, itemized quote for the complete repair of the existing unit. This quote must delineate the costs for:

- All required replacement components and materials.
- All necessary labor, listed by labor category and estimated hours.
- Any associated shipping, handling, or miscellaneous fees.
- The total cost to restore the unit to full operational capability.

Option B: Full Replacement Cost

A detailed, itemized quote for the purchase, delivery, and installation of a new, equivalent replacement unit. This quote must delineate the costs for:

- The new, equivalent replacement unit itself.
- Labor for the removal of the old unit and the installation and commissioning of the new unit.
- Any associated shipping, freight, and handling fees.
- The total cost to bring the system to full operational capability.

5.0 Special Regulations to Meet

All work shall comply with DAFMAN 91-203, DAFI 91-202, and all applicable federal, DoD, and Air Force safety, environmental, and occupational health requirements.

6.0 Qualifications/Certifications

The contractor shall ensure all personnel performing work under this contract are fully trained, certified, and qualified to maintain and repair FCX Systems equipment.

Documentation shall be provided upon request.